FERHAT BAŞ



Lead / Senior Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-stack engineer and technical lead, 25 years in. Most of that time went into clinical software for dialysis clinics, where a wrong lab value or a billing mistake reaches a patient the same day; that is where I learned to test properly and to look at the database before the screens. I like being on the whole arc of a product: sitting with the customer on requirements, modelling the data, drawing the screens, writing the code, and being the one who answers when something breaks at seven in the morning. I lead when a team needs it and I still write code every day; in my last role I was lead engineer and IT manager for the same company at the same time. New tools don't worry me, AI tools included, but I read what they produce before it ships.

CORE SKILLS

Frontend: React, Next.js, TypeScript, JavaScript (ES6+), jQuery, Redux / Context API, Tailwind, Bootstrap, PrimeReact, Blazor

Backend: C# / .NET, EF Core, Node.js, Express, PHP, Laravel, Symfony, Doctrine

Services and real-time: REST APIs, microservices, WebSocket, SignalR, SOAP / WSSE, HL7, TCP/IP, binary protocol parsing, FHIR R4

AI and ML: TensorFlow.js, PyTorch / LoRA fine-tuning, faster-whisper, local LLMs (node-llama-cpp), Gemini Live API, Claude, Antigravity

Databases: MSSQL, MySQL, MongoDB, Oracle, PostgreSQL, schema design, T-SQL, stored procedures

DevOps and tools: Docker, Azure, Azure DevOps, CI/CD, Git, Linux, Windows Server, VMware, Jira

Standards and compliance: ISO 27001, ISO 15504 (SPICE), healthcare regulations, Ministry of Health integrations

Also: Embedded C/C++ (ESP32 / ESP-IDF, NXP / Azure RTOS), Java

PROFESSIONAL EXPERIENCE

Lead Software Engineer & IT Manager — DMED Software *Apr 2023 – Jul 2026 · İzmir, Turkey*

- Led development and IT operations for the company's healthcare products; set the architecture, coding conventions and release process the team worked to.
- Ran the full product cycle: requirements with customers (Turkey, Qatar, North Macedonia), data modelling, UX mockups, implementation, release and production support.
- Delivered the North Macedonia D.Sys, RTMon, D.Sys HMS and the Card Management Platform (see Project Portfolio), working directly with an international client team in English.
- Owned servers, security, backups and government-compliance systems; accountable for the availability of every live system.

Lead Software Engineer & IT Manager — DMED Dialysis *Jul 2016 – Apr 2023 · İzmir, Turkey*

- Led a four-engineer development team and all IT operations for a clinical platform serving 19 clinics across 11 cities.
- Owned the full software, infrastructure and security lifecycle for organization-wide clinical and enterprise systems.
- Modernized legacy platforms while keeping DiaLINE live for 1.35M+ treatment sessions a year.
- Extended the platform from clinical records into workforce, finance and performance management used by 1,300 staff.
- Management Representative for the ISO 15504 (SPICE, from Apr 2018) and ISO 27001 (Nov 2018 – Feb 2019) certifications: wrote the policies and procedures, ran internal audits, and set up the Git and Redmine servers the processes run on.
- Scrum Master and developer on internal products (E-Exam, Document Management System); delivered D.Sys for a Qatar healthcare provider.

Software Developer — Diaverum · *Apr 2011 – Jul 2016 · İzmir* — Clinical and e-billing systems for dialysis centers, integrated with the Ministry of Health and laboratory systems; network and enterprise application availability.

Senior Software Developer — Dokuz Eylül University *Feb 2005 – Apr 2011 · İzmir, Turkey*

- Built the university hospital's dialysis clinical information system from the ground up, from requirements with clinicians to e-billing, laboratory and reporting integrations.
- Developed academic and healthcare web applications and database back-ends (Java, Oracle PL/SQL).

Web Developer *Before 2005*

- Built websites and web applications on small-scale projects.

EDUCATION & LANGUAGES

Business Administration & Management — Anadolu University (2001–2008)

Computer Programming — Dokuz Eylül University (1998–2001)

Languages: Turkish (native) · English (professional working proficiency)

PROJECT PORTFOLIO

Screens and module details for every product: ferhatbas.github.io

FLAGSHIP PROJECTS

D.Sys HMS — Global Hospital Information System *Apr 2025 – Jan 2026*

Country-neutral, multi-tenant HIS covering ADT, clinical orders/CDS, revenue cycle (billing, insurance claims, GL export, bank reconciliation) and FHIR R4 interoperability. Three independently deployable repos (API, web, DevOps) with an edge-adapter pattern isolating country-specific rules from the core.

Tech: *.NET 10 (custom DDD, EF Core, MSSQL, JWT), Next.js 16, React 19, PrimeReact 10, FHIR R4, Docker Compose*

Carbon HMS — Multi-Specialty Clinic Management Platform *2026 – present*

A generic healthcare application built to serve different disciplines on one shared core: hospital, polyclinic, private practice, dialysis, dental and aesthetics each get a specialty module on top of common patient, clinical, pharmacy, finance, document and multi-tenant identity modules.

Tech: *.NET 10, EF Core, MSSQL, JWT, React 19, PrimeReact, Tailwind, MongoDB, Docker, Azure*

D.Sys — Dialysis Information System *Nov 2020 – Aug 2024*

Hemodialysis information system in use in Qatar and North Macedonia: real-time monitoring, lab comparison with graphs, wound assessment, surveys and a multilingual UI. The North Macedonia version runs together with the RTMon platform below: built-in machine integration, SignalR live updates and a React-based RTMon Viewer.

Tech: *JavaScript (ES6), jQuery, WebSocket, SignalR, React, SOAP, MSSQL, Doctrine ORM, SCSS*

Dialysis Machine Integration Platform (RTMon) *Jan 2025 – Aug 2025*

Real-time patient monitoring. A TCP server ingests binary data from multiple machine brands, logs it to MongoDB and streams live parameters to the web app over WebSocket with token-based auth. Includes per-machine parameter management.

Tech: *C#, React, TCP/IP, WebSocket, MongoDB*

DiaLINE — Clinical and Enterprise Management Platform *Nov 2010 – Nov 2023*

Modules for dialysis clinics on one platform: **Clinical** (patient records, treatment, Ministry of Health and e-Nabız integration, 7 laboratory systems, e-prescription signing), **E-billing and revenue** (SGK billing), **Workforce management** (HR records, fingerprint-based work-hour tracking, shift planning, salary calculation), **Performance** (KPI management, productivity reporting) and **Event tracking**.

Scale (at peak): 53 clinics · 19 cities · 9,000+ patients · 1,300 staff · 1,350,000+ sessions / year

Tech: *PHP, MySQL, JavaScript, jQuery, SOAP, WSSE, C# (batch), Java (XAdES-BES signer)*

Card Management Platform *Apr 2023 – Jan 2025*

Smart-card management web application with card-reader integration and real-time updates.

Tech: *Blazor, C#, EF Core, SignalR, RFID reader/writer*

Document Management System *Nov 2021 – Nov 2023*

File management for healthcare companies: authenticated access, versioning, in-browser Office/PDF viewers, document templates and in-app messaging.

Tech: *React, Node.js, Express, MongoDB, React Native*

FS-SC — MCP-coordinated software company *2026 – present · Personal*

An MCP server that runs a 12-role software team (PM, PO, architect, backend, frontend, QA, E2E tester, DevOps, reporter, documentor, SCM, scrum master) through a 10-phase cycle from PRD and architecture to QA, E2E tests, DevOps and documentation. Every delivery is verified (syntax, install, lint, build, tests), failures open fix rounds and passing work is committed per Jira item; a live dashboard shows the Jira tree, Kanban, agent mesh and sprint reports.

Tech: *Python, FastAPI, MCP SDK, WebSocket, React Flow, Tailwind, Playwright, pytest*

AI & MACHINE LEARNING

MedDikte

Offline medical dictation: speech-to-text, four-layer hallucination shield, LoRA fine-tuning pipeline with WER/MER evaluation

Tech: *Tauri (Rust), Python/FastAPI, faster-whisper, PyTorch/LoRA, React, SQLite*

Anemia-API

In-house ML model that recommends ESA and iron dosage from hemoglobin and iron levels. Training and serving in one service: data processing, model training and a documented prediction API

Tech: *Node.js, TypeScript, MongoDB 8, TensorFlow.js, OpenAPI, Docker*

Clinical Voice Assistant

Turkish voice-command assistant embedded in the DiaLINE clinical decision-support system (in production)

EMBEDDED & IOT

Embedded / IoT *2021 – 2025*

ESP32 and NXP i.MX RT device gateways with HTTP/WebSocket servers, TLS and smart-card integration.

Tech: *C/C++, ESP-IDF, FreeRTOS, Azure RTOS / ThreadX*

EARLIER WORK

E-Exam *(from 2018)* Employee learning and assessment platform: managers create and publish exams by employee type, one-time, at hiring or periodically, and track completion. *PHP, JavaScript, REST microservices, Mocha*

E-Signature Application *(2019)* Manages user token devices and signs documents over WebSocket, using the TÜBİTAK API and MS Winscard API. *C#, PHP, JavaScript*

Nefroliz *(2003 – 2010)* Dialysis clinical information system with hemodialysis, peritoneal dialysis and transplantation modules, e-billing, laboratory integration and quality-indicator reports. *C++ Builder, MySQL*

Medical Device Integration V2 *(2007 – 2008)* Template-driven communicator for blood-analysis devices over RS232 or TCP/IP. *C#, Oracle PL/SQL*

HL7 Communication Application *(2005 – 2006)* Transfers blood-center data to the HIS in HL7 format. *Java, Oracle PL/SQL*